

H2ANDERSON SERANGOON JUNIOR COLLEGE
HIGHER 2**2021 JC2 PRELIMINARY EXAMINATION**CANDIDATE
NAME**MARK SCHEME**

CLASS

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INDEX
NUMBER

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H2 BIOLOGY**9744/03**

Paper 3 Long Structured and Free-Response Questions

30 August 2021

Candidates answer on Question Paper.

No additional Materials are required

2 hours**READ THESE INSTRUCTIONS FIRST**

Write your name, class and index number in the spaces at the top of this page.

Write in dark blue or black pen.

You may use a HB pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section AAnswer **all** questions in the spaces provided on the Question Paper.**Section B**Answer any **one** question in the spaces provided on the Question Paper

For Examiner's Use	
1	
2	
3	
4/5	
/75	

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

Section A

Answer **all** the questions in this section.

- 1 Development and function of an organism is largely controlled by genes. Any accidental changes to an organism's genes, in the form of gene mutations or chromosomal aberrations, can be potentially damaging.

(a) Compare gene mutation and chromosomal aberration.

Similarities *[max 2]*

1. Both may lead to a **change in phenotype**.
2. Both result from **errors that occur during DNA replication/** are caused by **mutagens**

Differences *[max 2, must be PTPC]*

3. A gene mutation involves a permanent **change** in DNA **nucleotide/ base sequence** but a chromosomal aberration involves a change in the **number** or **structure** of chromosome(s).
4. A gene mutation is brought about by base-pair **insertion, deletion** or **substitution**, but a chromosomal aberration is caused by **reshuffling of existing alleles** on a chromosome (structural aberration)
5. or **non-disjunction** of chromosomes during meiosis or mitosis (numerical aberration).
6. Only **one gene/ gene locus** is affected in a gene mutation, while **multiple genes/ loci** are affected in chromosomal aberration.

[4]

Gene mutations can also affect haemoglobin molecule, leading to sickle cell anaemia. Hb^A codes for the normal functional haemoglobin while Hb^S is the mutant recessive allele for sickle cell anaemia.

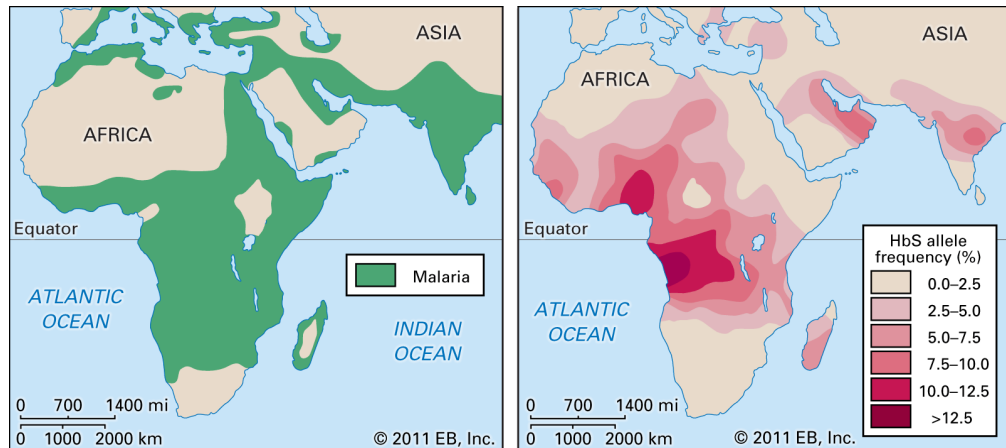


Fig. 1.1

- (b) Comment on whether the data provided in Fig. 1.1 support a correlation between Hb^S allele frequency and incidences of malaria.

Yes

- Hb^S maintained at high frequency in areas where malaria is widespread in **Africa** and **southern Asia**;
- **Hb^A Hb^S** is at a **selective advantage** because their red blood cells sickle only under conditions of very low oxygen levels such as at high altitude/ The parasite is less adapted to living in red blood cells half of whose haemoglobin is Hb^S.

No

- Regions with *no malaria* but there are incidences of Hb^S in **northwest Africa** - Hb^A Hb^S only sickle cell **trait**, **no symptoms** of sickle cell disease unless at high altitudes/ live a normal life/ Hb^A Hb^S is a **carrier** - Can mate and pass on to offspring.
- Regions with *malaria* but *no Hb^S* in **southern Africa** and **southern Asia** – Malaria **worsens** the pre-existing anaemia and causes Hb^S Hb^S individuals to die, **cannot pass on Hb^S allele** to offspring.

[2]

Malaria is caused by the *Plasmodium* parasite. It has developed a wide range of defences against the human immune system, making an effective vaccine difficult to develop.

Randomised trials of a new vaccine called RTS were conducted in 2011. 894 children aged 5 to 17 months from Kenya and Tanzania were vaccinated with either the RTS vaccine or with a placebo (no active substance). Neither the children nor their families were told whether they were given the vaccine or a placebo.

The children were then monitored. Among those who were vaccinated, 40% fewer children developed malaria.

- (c) Describe the role of the placebo.

[1]

(i)

The placebo serves as a negative **control** to show that the 40% **lesser** children who developed malaria is due to the effectiveness of the vaccine and not due to any other variables.

(ii) Discuss the **economic** benefits of vaccination against malaria. [3]

1. Herd immunity/ Vaccinated individuals have lesser chance of getting malaria, therefore can be economically productive/ protection of those who cannot be immunized.
2. Reduce strain on healthcare system and reduce healthcare cost, more resources for other diseases,
3. When Malaria can be eradicated, the country can develop its tourism sector further and the country can also attract more investments. [3]

(iii) One way to produce immunity to malaria in humans is to inject a vaccine consisting of irradiated malarial parasites that are unable to divide.

Assuming malaria parasites divide similarly to human cells, explain how the irradiation changes the pathogen so that it is unable to divide.

- Ionising radiation e.g. X-rays / gamma rays **causes damage to DNA/ mutations**
- Changes shape of **proteins** eg. **enzymes** for organelle synthesis, **spindle fibres** not formed;
- **Cell cycle checkpoints** are inactivated/ G_1 not able to proceed to S and G_2 even when conditions are met eg. adequate cell size, sufficient nutrients and growth factors; [2]

In another study to test the ability of the RTS vaccine to protect against malaria, a clinical trial was conducted where recruited patients for the test were randomly given either the vaccine or a placebo. They were then tested for antibodies against the malaria parasite after 10 weeks.

Table 1.1 shows the results of the trial.

Table 1.1

test group	no. of patients	average amount of antibodies / arbitrary units
vaccine	10	3148
placebo	10	489

standard deviation $s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$

t -test $t = \frac{|\bar{x}_1 - \bar{x}_2|}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}} \quad v = n_1 + n_2 - 2$

Symbols:

x = observed value

$\bar{x}$ = mean

n = number of observations (i.e. number of values within a data set)

v = degrees of freedom

Table of t critical values

degrees of freedom	probability, p				
	0.5	0.1	0.05	0.01	0.001
1	1.00	6.31	12.71	63.66	636.62
2	0.82	2.92	4.30	9.92	31.60
3	0.76	2.35	3.18	5.84	12.92
4	0.74	2.13	2.78	4.60	8.61
5	0.73	2.02	2.57	4.03	6.87
6	0.72	1.94	2.45	3.71	5.96
7	0.71	1.89	2.36	3.50	5.41
8	0.71	1.86	2.31	3.36	5.04
9	0.70	1.83	2.26	3.25	4.78
10	0.70	1.81	2.23	3.17	4.59
11	0.70	1.80	2.20	3.11	4.44
12	0.70	1.78	2.18	3.05	4.32
13	0.69	1.77	2.16	3.01	4.22
14	0.69	1.76	2.14	2.98	4.14
15	0.69	1.75	2.13	2.95	4.07
16	0.69	1.75	2.12	2.92	4.01
17	0.69	1.74	2.11	2.90	3.97
18	0.69	1.73	2.10	2.88	3.92
19	0.69	1.73	2.09	2.86	3.88
20	0.69	1.72	2.09	2.85	3.85

(iv) The calculated t value is 2.365.

Based on the information provided, explain if the vaccine is effective in conferring protection against malaria. [3]

1. At **degrees of freedom of 18**, the calculated t-value of 2.365 is **greater** than t-critical value of 2.10 at $p=0.05$ OR $p < 0.05$
2. there is **significant difference** between patients given the vaccine and patients given the placebo and **any difference is not due to chance**.
3. Hence, the vaccine is **effective** as it resulted in significantly higher amounts of antibodies against *Salmonella*;

For individuals who are carriers of sickle cell anaemia allele, the best way to find out how sickle cell disease runs in their family is for a genetic counsellor to do a genetic test. Genetic information of the beta-globin alleles can be obtained through DNA sequencing, which is a modified Polymerase Chain Reaction (PCR).

DNA is extracted from the individual to be tested and used as a template in four independent PCR reactions. Each reaction contains the standard PCR components with dNTPs (deoxyribonucleoside triphosphates), and a different ddN (dideoxynucleoside triphosphate): ddA, ddG, ddC or ddT. Random incorporation of the ddN into the growing DNA chain results in chain termination. This generates a range of DNA strands of different lengths.

The DNA strands from each reaction are then **separated** and visualised using a DNA-staining dye. Each band corresponds to a DNA strand that has incorporated the ddNTP.

Fig. 1.2 shows how DNA sequencing can be carried out.

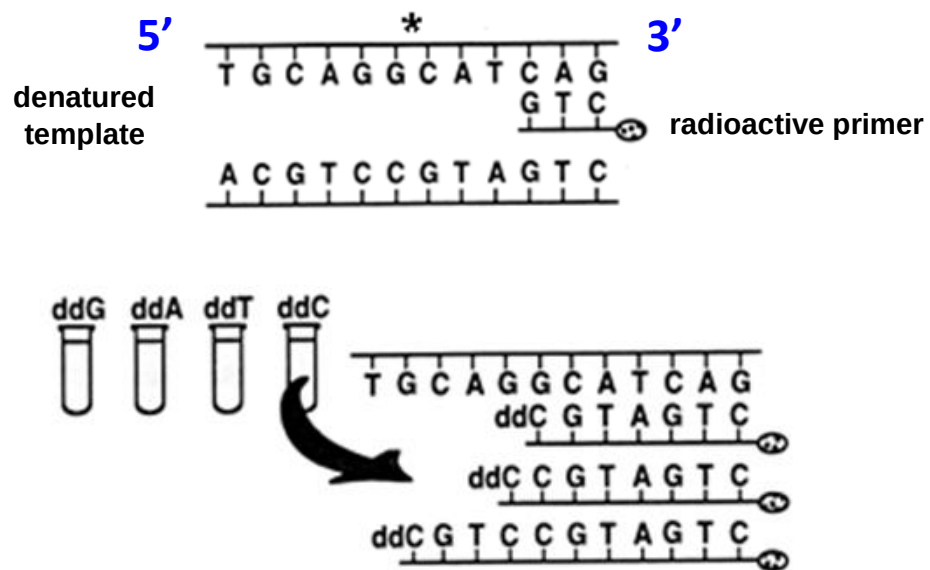


Fig. 1.2

Fig. 1.3 shows the structure of a ddNTP.

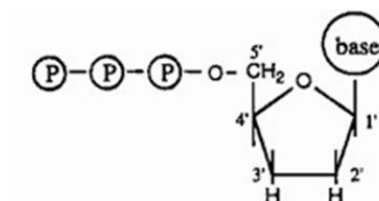


Fig. 1.3

The second procedure of the sequencing process produces a result shown in Fig. 1.4, from which the DNA sequence can be read.

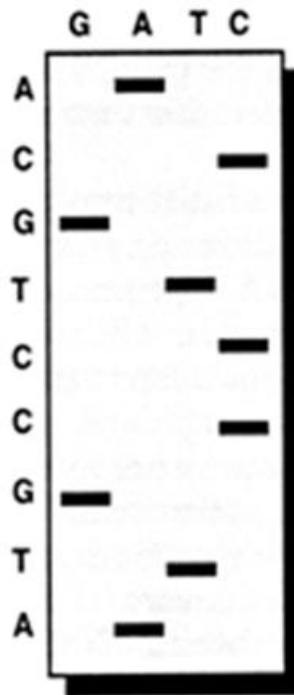


Fig. 1.4

- (g) (i) On Fig. 1.2, label the 5' and 3' ends of the DNA strand marked with an asterisk (*). [1]
- (ii) With reference to Fig. 1.3, explain how ddNTP results in the termination of chain elongation.
 1. ddNTP has no 3'-OH* group;
 2. 5' phosphate group on incoming deoxyribonucleoside triphosphate/ deoxyribonucleotide cannot form phosphodiester bond* with the ddNTP;
 OR
 DNA/Taq polymerase active site* is complementary in shape and charge to 3'-OH group, which is necessary for chain extension; [2]
- (iii) With reference to Fig. 1.4, state the sequence of the longest strand synthesised. Indicate the 5' and 3' ends of the strand in your answer.
 3' – ACGTCCGTA – 5' OR 5' – ATGCCTGCA – 3' [1]
- (iv) Describe the procedure used to separate the DNA strands in Fig. 1.4.
 1. **Gel electrophoresis**
 2. An **agarose gel** is submerged in a **buffer solution** containing ions that will **conduct electricity**.
 3. DNA fragments are loaded into **wells** close to the **cathode / negative electrode**.
 4. **Loading dye** contains glycerol which is heavy for DNA to sink.
 5. A **direct current** is applied.
 6. **Negatively charged DNA** moves toward the **anode / positive electrode**.
 7. Shorter DNA fragments move **faster** and further than longer ones. [4]

[Total: 23 marks]

- 2 Fig. 2.1 shows the exchange of substances between a mitochondrion and a chloroplast in a leaf cell.

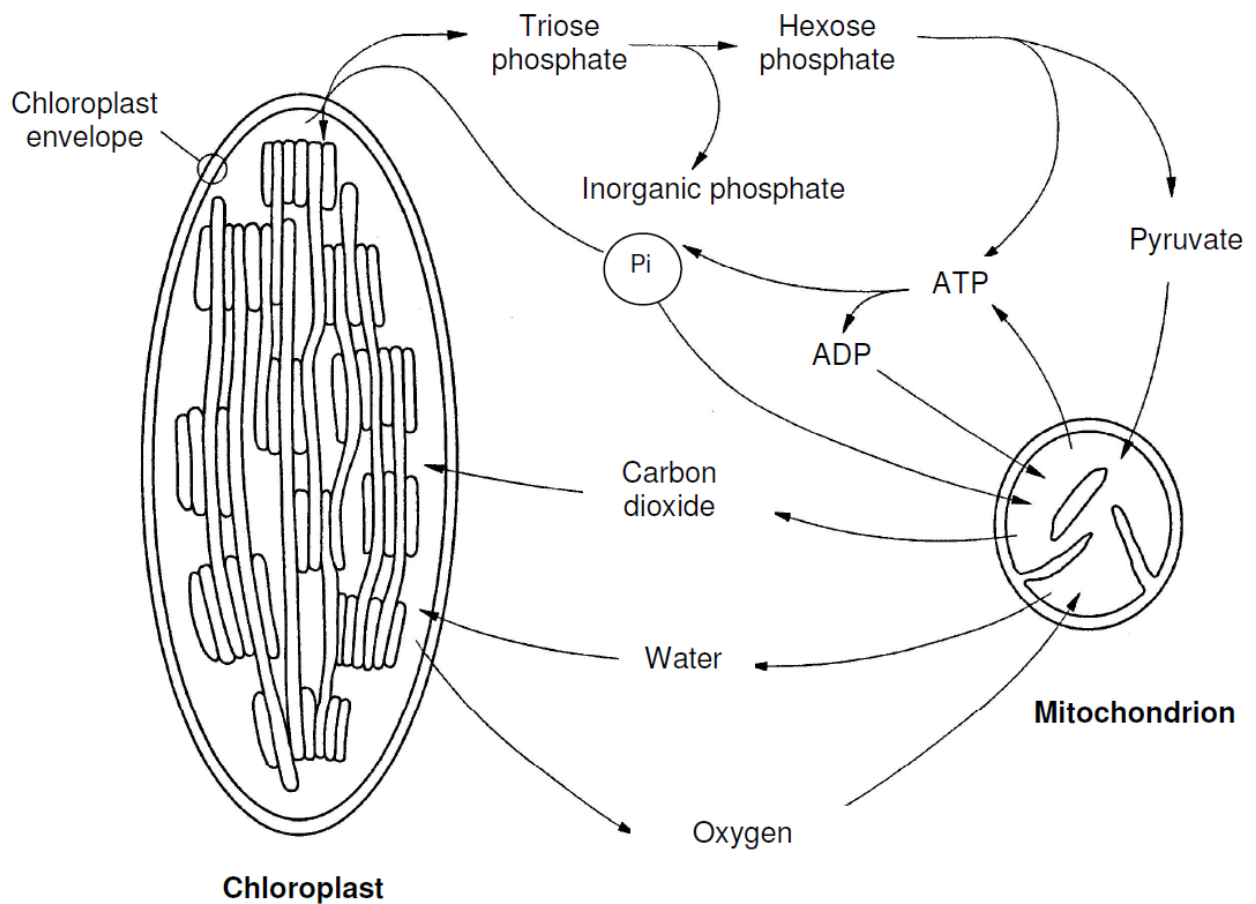


Fig. 2.1

- (a) Explain how triose phosphate and hexose phosphate link the metabolic pathways between the chloroplast and the mitochondrion of this cell. [3]
1. Triose phosphate is formed as a **product** of the **light-independent reactions/ Calvin cycle/** when **PGA is reduced** in the **stroma** of the chloroplast
(*triose phosphate exits the chloroplast through carrier proteins present in the chloroplast envelope*)
 2. **two** triose phosphate (3C) molecules **combine** to form **one** hexose phosphate (6C) in the **cytosol**
 3. that can **enter glycolysis** as a **respiratory substrate/ intermediate**, to be **broken down to form pyruvate** (3C)
 4. pyruvate then **enters the mitochondrial matrix** to undergo the **link reaction/** be **converted to acetyl-coA**

(b) ATP is produced in both the chloroplast and the mitochondrion.

- (i) Complete the table below by listing **all** the types of phosphorylation reactions occurring within these organelles that produce ATP. [1]

organelle	type(s) of phosphorylation reaction occurring within
chloroplast	
mitochondrion	

chloroplast **photophosphorylation**

mitochondrion **substrate-level** phosphorylation and **oxidative** phosphorylation

- (ii) Outline the fates of the ATP molecules produced in these organelles. [3]

[in chloroplast]

1. ATP (produced during photophosphorylation/ in the light-dependent stage) is involved in the **phosphorylation** of **PGA/** phosphoglyceric acid to **triose phosphate** (Accept: *glyceraldehyde-3-phosphate/ G3P*) in the **carbon/ PGA reduction phase** of the **Calvin cycle**
2. ATP is also the source of **energy** in **RuBP/ ribulose biphosphate regeneration** during the **Calvin cycle**

[in mitochondrion]

3. ATP (produced during substrate-level/ oxidative phosphorylation) **leaves the mitochondrion (OWTTE)** and is used in **cellular activities that require energy** or give named example, e.g. active transport, movement of vesicles, amino acid activation, ATP used in energy investment phase of glycolysis.

- (iii) An aqueous solution contains small membranous vesicles which are derived from a mitochondrial membrane. In these vesicles, the orientation of the ATP synthase enzymes is preserved as it is in the cell.

Fig. 2.2 shows four experimental systems **A**, **B**, **C** and **D** that are prepared in an attempt to synthesise ATP.

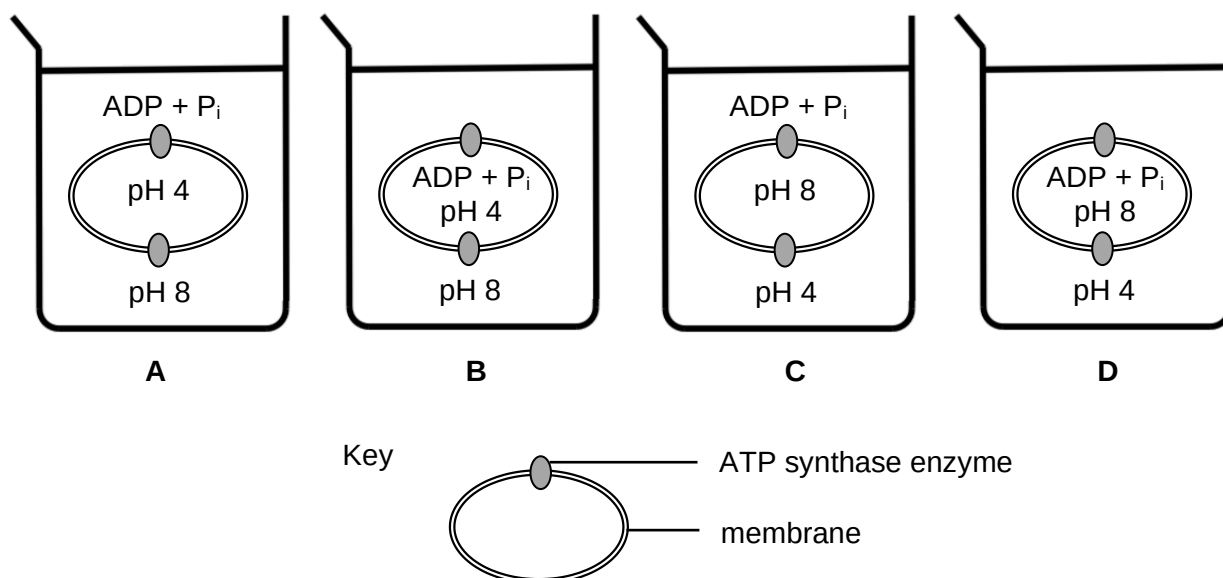


Fig. 2.2

Identify the experimental system that will result in the successful synthesis of ATP, and explain your choice. [3]

experimental system

explanation

1. experimental system **D**

[any 2]

2. There is **higher H⁺ concentration outside the vesicle** (as indicated by pH4, similar to inter-membrane space)/ **lower H⁺ concentration inside the vesicle** (as indicated by pH8, similar to mitochondrial matrix)
3. A **proton gradient** (Accept: *proton motive force*) is thus correctly established across the vesicle membrane in D that mimics that found across the inner mitochondrial membrane. This causes H⁺ ions to **diffuse down their gradient into the interior of the vesicle** via ATP synthase
4. which synthesises ATP, using the ADP and P_i which have to be within the vesicle as they are the **raw materials/ substrates** used by ATP synthase

Hexokinase is the enzyme involved in the first step of respiration. It catalyses the formation of G6P (glucose-6-phosphate).

ASR scientists recently invented inhibitor **X**, which can affect hexokinase activity.

An investigation was carried out to determine the effect on the production of G6P when the concentration of inhibitor **X** is increased. Five different concentrations (0 μM to 30 μM) of the inhibitor were used.

The concentration of G6P produced in each reaction mixture was determined at 10 minute intervals, up to 60 minutes.

The results are shown in Fig. 2.3.

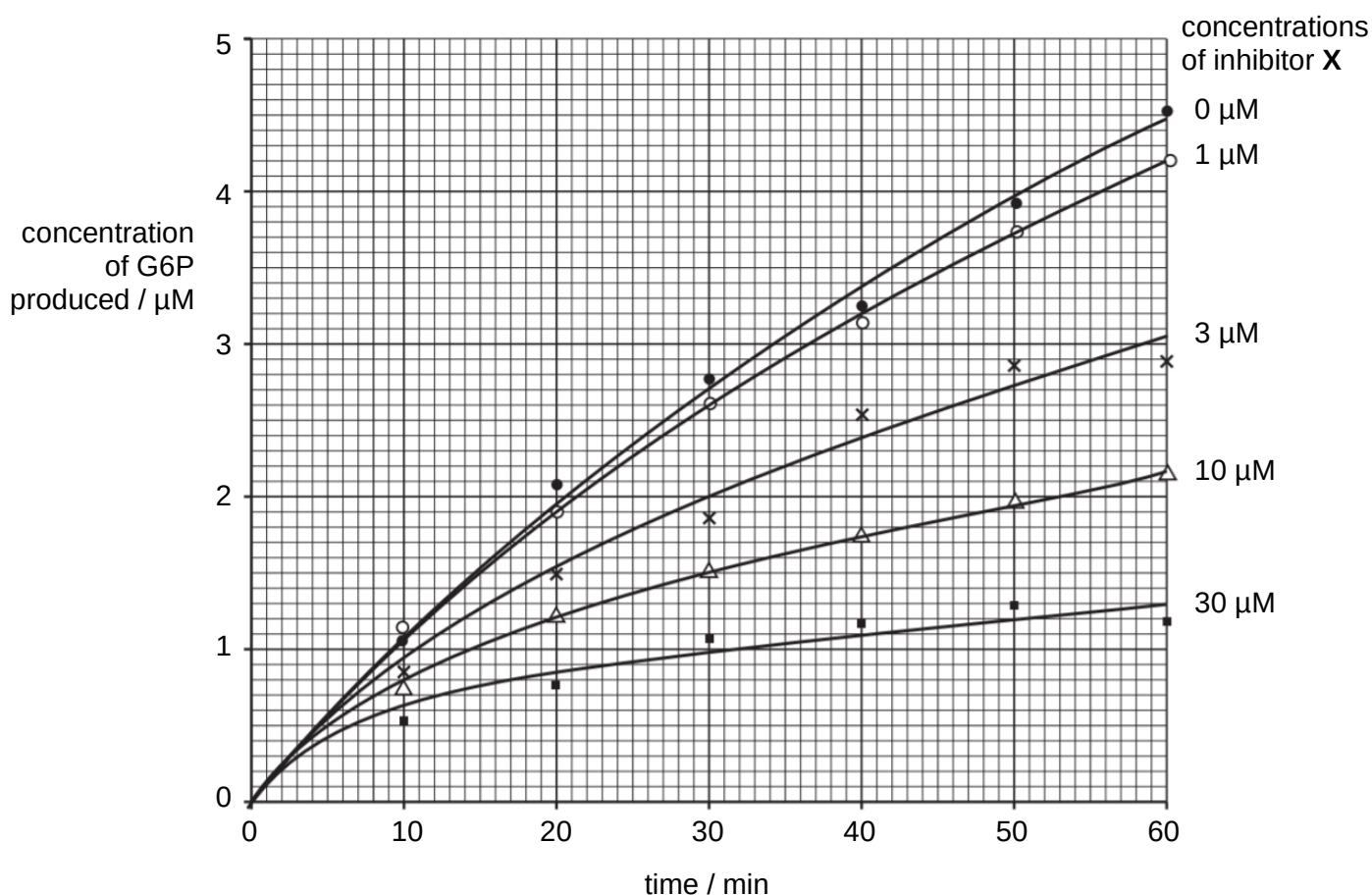


Fig. 2.3

(c) Use Fig. 2.3 to describe the effect of inhibitor **X** on the **rate** of G6P produced during the duration of the investigation. [3]

[any 3]

1. From **0 to 2** (Accept: 2.5/ 3) **minutes**, inhibitor X has **no noticeable effect** (OWTTE) (Accept: no effect) as there is **no difference in rate of G6P produced/ same rate**, for all concentrations of inhibitor AND the reaction without inhibitor
2. From **2** (Accept: 2 – 5) **minutes onwards**, the inhibitor X **decreases/ slows the rate of G6P produced/** rate becomes **less steep** for all/ most concentrations of inhibitor COMPARED TO the uninhibited reaction.
3. **Increasing the concentration of inhibitor X** leads to a **decrease/ slows the rate of G6P produced**

4. AVP, e.g. 1 μM of inhibitor had minimal effect on rate of G6P produced

Note: rate is indicated by the gradient of the various curves

- (d) From this investigation, it is **not** possible to conclude whether inhibitor **X** functions in a competitive or non-competitive manner.

Comment on the additional information needed to make a conclusion on how inhibitor **X** functions. [2]

[any 2]

1. Effect on X on rate of reaction at **low and high substrate (glucose) concentrations/ different substrate concentrations**, compared to the same range of substrate concentrations without X
2. If **maximum rate** of reaction/ V_{max} of the uninhibited reaction is **achieved by the inhibited reaction at high substrate concentrations/** if the **effect of the inhibitor decreases** with an **increase in substrate concentration**, X is a **competitive inhibitor** (or reverse argument)
3. AVP, e.g. if **shape** of X is **similar to substrate/ glucose**, it could be a **competitive inhibitor/** if **shape** of X is **complementary** to an **allosteric site** on hexokinase, it could be a **non-competitive inhibitor**.

[Total: 15]

- 3 *Salmonella enterica* is a species of pathogenic bacteria that causes gastrointestinal disease in humans and animals. The pathogen is shown to be remarkably adaptive, being able to invade a large range of host organisms.

Salmonella enterica serotype *typhimurium* (*S. typhimurium*) is the most significant cause of food poisoning in western countries. Infection is possible from ingestion of contaminated food and water. *S. typhimurium* typically enters the body via the gut. The intestinal barrier thus plays an important role in preventing infections by *S. typhimurium*.

Fig. 3.1 shows the composition of the intestinal barrier.

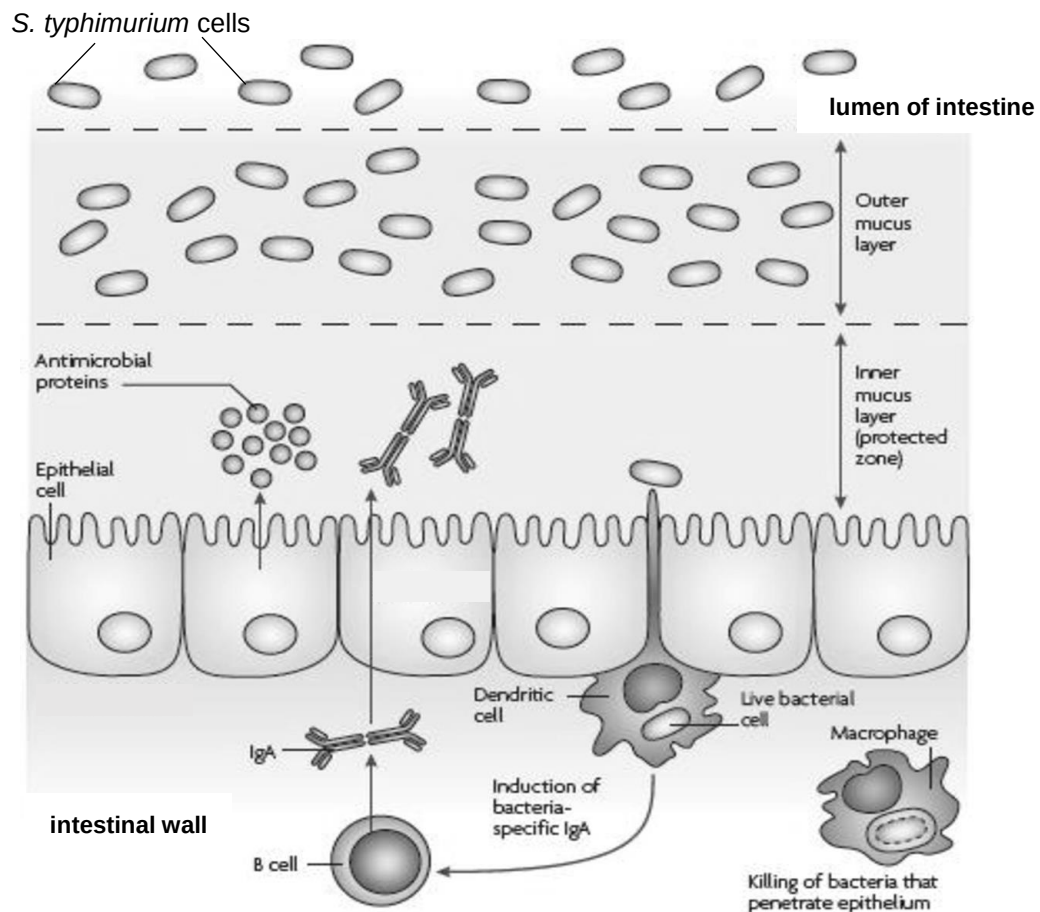


Fig. 3.1

- (a) With reference to Fig. 3.1 and biological knowledge of the human immune system,
- (i) describe the role played by the gut epithelial cells in preventing infections of *S. typhimurium*. [3]

Any 3

1. Epithelial cells form a layer, serving as an **anatomical / physical barrier** to **prevent entry** of *S. typhimurium* into the body;
2. Epithelial cells **secrete mucus** forming a layer on epithelial cells, preventing penetration of *S. typhimurium* by **trapping bacteria**;
3. **Presence of cilia** on the epithelial cells that help to **sweep the mucus** containing *S. typhimurium* **away for removal**.
4. Epithelial cells **secrete antimicrobial proteins** into inner mucus layer which help to disrupt the cell membrane of *S. typhimurium* and **kill S. typhimurium**
5. Epithelial cells secrete **lysozyme** into inner mucus layer that **degrades peptidoglycan cell walls** of *S. typhimurium* (idea of breaking down cell wall), leading to lysis

(ii) outline how antibodies against *S. typhimurium* are produced. [4]

Any 4

1. **Dendritic cells** act as antigen presenting cells (APCs) and **engulf** *S. typhimurium* bacteria cells via **phagocytosis**
2. DCs present processed antigen on **cell surface** in the form of **peptide:MHC (Class II) complexes** that are recognised and bound by the **specific T cell receptor** on **naïve T cell** that is **complementary** in shape
3. **Activated T cells** undergo **clonal expansion (or divide)** and **differentiation** to form **helper T cells** (and cytotoxic T cells)
4. **Helper T cells** **secrete cytokines** that **activate** specific **naïve B cells** to (undergo clonal expansion and) differentiate into plasma cells
5. **Plasma cells secrete antibodies**
6. **Activated B cells** undergo **class switching** occurs to result in production of *S. typhimurium*-specific **IgA**

- (b) The lumen of the intestinal tract contains nutrients and supports the growth of a diverse community of healthy bacteria that makes up the microbiota.

Suggest how the microbiota found in the lumen and in the outer mucus layer could help to protect against infection by this pathogenic bacteria. [1]

Any 1

1. **Compete with pathogenic bacteria for nutrients** to prevent their growth in intestinal lumen;
2. **Physical barrier to prevent the attachment of bacteria** to epithelial cells lining lumen (idea of blocking);
3. **Produce compounds/toxins that inhibit growth** of pathogenic bacteria;

A: any reasonable suggestion on how existing bacteria may kill or inhibit growth of pathogenic bacteria

A novel *Salmonella* vaccine was developed by using a strain of *Salmonella* that has been rendered harmless. It is believed that newborns can be protected from *Salmonella* infections when their mothers are vaccinated during their pregnancy.

- (c) Explain how vaccination of pregnant woman can confer protection against *Salmonella* infections in newborns. [2]

1. **Maternal antibodies against *Salmonella* bacteria can be passed on** to the newborn via the **placenta or breast milk** (R: umbilical cord/breastfeeding)
2. **Ref to presence of antibodies that are immediately available** for use in the newborns
3. **Ref to explanation of specific effector role of antibodies:** e.g. antibodies bind to the toxins of *Salmonella* bacteria to prevent its harmful effect (neutralisation) A: opsonisation/agglutination

- (d) Explain why children need to be revaccinated although their mothers have been vaccinated during pregnancy. [2]

1. The protection via the mother is **transient/short-term**
2. as maternal antibodies will be **eventually degraded**
3. Vaccination of children allows for the development of **memory cells/immunological memory** that allows the child to respond to any further infection in its lifetime.

[Total:12]

Section B

Answer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answer must be set out in parts **(a)**, **(b)**, etc., as indicated in the question.

- 4 Membranes and nucleic acids are commonly found in all living organisms.

(a) Discuss the importance of membranes to biological processes in the cell. [12]

(b) Prokaryotes and eukaryotes maintain the integrity of their genetic material to ensure that changes are minimal.

Outline why genetic stability is necessary in living organisms, and explain how genetic stability is maintained at the molecular, cellular and population levels. [13]

- 5 The cell is the basic unit of life, within which numerous biological processes that are essential to life occur.

(a) Discuss the importance of forming bonds that are weak and temporary during biological processes in the cell. [12]

(b) Cells are able to divide and give rise to new cells.

Cyclins are an important family of proteins that control the rate of cell cycle progression of a dividing cell. They are so named because, in general, the concentrations of the various cyclins change in a cyclical manner during the cell cycle. The changing levels of active cyclins thus ensure that there is sufficient time for various checks to be performed, to ensure that key cell cycle events have occurred normally.

Outline the checks made at cell cycle checkpoints, and explain how a dividing cell is able to **quickly** regulate the levels of cyclins which in turn control these checkpoints. [13]

[Total: 25]

QUESTION 4

Membranes and nucleic acids are commonly found in all living organisms.

(a) Discuss the importance of membranes to biological processes in the cell. [12]

Transport Processes (within the cell and across the membrane) (max 6)

- T1. Membranes are selectively permeable, membrane only allows some molecules to cross directly without assistance
- T2. Allows for formation of unique environment for highly specialised activities (idea of compartmentalisation)
- T3. Ref to only allowing nonpolar molecules to diffuse across freely OR charged and polar molecules are repelled by hydrophobic core require presence of transport proteins embedded
- T4. Channel protein / carrier proteins embedded allow for facilitated diffusion of specific molecules across cell surface membrane into or out of cell down a concentration gradient;
- T5. Example: aquaporins allow for water to move into or out of cells / glucose transporters allow for uptake of glucose into cells for respiration;
- T6. Pumps embedded in cell surface membrane allow for active transport of molecules into or out of cells against a concentration gradient;
- T7. Example: sodium-potassium pump which pumps three sodium ions and two potassium ions out and into cells respectively to maintain membrane potential;
- T8. Fluidity of phospholipid bilayer allows formation of membrane pores that allow specific molecules to be transported across the cell membrane
- T9. Example: Nuclear pores allow nucleotides to enter nucleus for DNA replication / transcription
- T10. Fluid membrane also allows for budding/invagination (or inward folding) to form vesicles/vacuoles for endocytosis/protein secretion OR fusion of vesicles **A: any named process involving fusion of vesicles or budding**
- T11. Ribosomes, which are embedded onto membranes such as cisternae of the rough endoplasmic reticulum, which synthesise polypeptides;
- T12. Protein packaged into transport vesicles which bud off from membrane of rough ER and move to Golgi apparatus
- T13. packaged into secretory vesicles which bud off from trans face of Golgi apparatus and move towards cell surface membrane for secretion out of the cell;
- T14. Post-translational modification occurs to insulin within the cisternae/lumen of rough ER / within cisternae of Golgi apparatus

Cell Signalling / Communication

- C1. Receptors are embedded on cell surface membrane for cell to cell communication/recognition; **[award once] A: Any other relevant examples such as glucagon receptor**
- C2. Receptors bind to ligand/signal molecule to lead to downstream transduction and cellular response
- C3. Example: for homeostasis/regulation of blood glucose level/regulate gene expression (rate of transcription of target gene)
- C4. Glycoproteins embedded on cell surface membranes of epithelial cells for cell-to-cell adhesion

- C5. *Ref to presence of enzyme on the cell surface membrane (i.e. adeny cyclase) for production of second messenger (i.e. cAMP) to relay the signal/trigger phosphorylation cascade*

Immune Response [max 4]

- I1. Receptors are embedded on cell surface membrane for **cell to cell communication/recognition** [award once]
- I2. allows for the uptake of antigens via formation of pseudopodia (or via receptor-mediated endocytosis) into APCs to be processed (**note: receptor-mediated endocytosis, not 'cell-mediated endocytosis'**)
- I3. for antigen of pathogen to be presented on cell surface membrane of APC (i.e. macrophage, dendritic cell/B cell)
- I4. Ref to **B/T cell receptor** (TCR) embedded on cell surface membrane of **B/T helper/cytotoxic cells**
- I5. B cell receptor (BCR) embedded on cell surface membrane of naïve B cells to recognise **specific epitope/antigen** on pathogen OR
- I6. TCR binds to peptide-MHC complex on naïve B cells to activate naïve B cells into antibody-producing plasma cells/ activation of naïve T cells / activation of memory B cells

Photosynthesis and Respiration [max 6]

- PR1. Allows **functionally-related proteins** to be **grouped** together for sequential biochemical processes
- PR2. Ref to **photosystems, electron transport chains** and **ATP synthases embedded** on thylakoid membrane/(cristae of) inner mitochondrial membrane (IMM)
- PR3. Photosystems/photosynthetic pigments embedded allow for absorption of light energy during the light dependent stage of photosynthesis / photophosphorylation;
- PR4. Inner mitochondrial membrane is **highly folded** into cristae to increase surface area to hold more proteins
- PR5. **Energy released** (**note: released, not produced**) during the **movement of electrons** down embedded **electron transport chain** results in the **formation of a proton gradient/active transport (or pumping) of protons** across the thylakoid membrane/IMM;
- PR6. Hydrophobic core of thylakoid membrane/IMM is **impermeable to H⁺** (or idea of presence of 'space' between two membranes that allows for accumulation of protons) and hence allows for **proton gradient** to be generated
- PR7. ATP synthase complexes allow diffusion of protons through the hydrophilic channel that generate proton motive force to synthesise ATP via chemiosmosis;
- PR8. *AVP (e.g. presence of cellulose synthase to catalyse the synthesis of β -glucose to form cellulose cell wall)*

QWC: scope of answer to include roles of membranes from three different named biological processes

- (b) Prokaryotes and eukaryotes maintain the integrity of their genetic material to ensure that changes are minimal.

Outline why genetic stability is necessary in living organisms, and explain how genetic stability is maintained at the molecular, cellular and population levels. [13]

Part I: Outline why genetic stability is necessary in living organisms

- N1. Produce **genetically identical** cells for **growth of organism**
- N2. For **replacement of worn out/damaged** cells (or tissue repair)
- N3. **Ref to method of reproduction** in organisms that undergo asexual reproduction/prokaryotes
- N4. **Ref to rapid colonisation of a habitat** during times of constant environment (or no major envt changes)
- N5. To ensure **survival** of species to **reproductive age**
- N6. Give rise to subsequent **generations** to ensure that species **does not undergo extinction**
- N7. **Mutations** in (**named**) genes (ie. Haemoglobin gene or proto-oncogene) can cause diseases eg. sickle cell anaemia/ cancer.

Part II: Describe how genetic stability is maintained (max 9)

Molecular level (max 4)

- M1. Genetic stability at molecular level means maintaining same DNA sequence in daughter cells and parental cells;
- M2. In order to have identical copies of DNA as parent cell before nuclear division, DNA is first replicated during **S phase of interphase** via semi-conservative replication
- M3. Each parental strand of double stranded DNA act as template
- M4. To synthesise daughter strand via complementary base pairing;
- M5. Where nitrogenous base adenine forms 2 hydrogen bonds with thymine, guanine forms 3 hydrogen bonds with cytosine;
- M6. Each DNA molecule formed consists of one parental template with a daughter strand/newly synthesised strand;
- M7. DNA polymerase "proof-reads" newly synthesised region where if there is an incorrect DNA nucleotide is added, it will swiftly remove and replace with correct nucleotide. This is to ensure fidelity of DNA sequence;
- M8. Double stranded so that one of the strands can be a backup to be used as a template for repair
- M9. Numerous hydrogen bonds of DNA/ phosphodiester bonds between sugar-phosphate backbone increases stability of the molecule;
- M10. Telomeres at the ends of chromosomes prevent the loss of genetic information with each round of DNA replication;

Cellular level (max 4)

- C1. Genetic stability at cellular level means maintaining same number and type of chromosomes in daughter and parental cells
- C2. During prophase, identical DNA molecules condense to form genetically identical sister chromatids to allow for equal separation;
- C3. During metaphase, chromosomes align along metaphase plate/equator
- C4. During anaphase, centromere divides and kinetochore microtubules from either pole attached to each sister chromatids, shortens;
- C5. During anaphase, sister chromatids are separated and move to opposite poles R: opposite ends

- C6. During telophase/cytokinesis **A: mitosis**, formation of nuclear envelope/cell divides to ensure equal distribution of chromosomes to daughter nuclei/cells
- C7. Cell cycle is regulated at G₁, G₂ and M checkpoints to ensure the previous stage has occurred correctly before moving onto the next stage
- C8. G₁ and/or G₂ checkpoints checks for **DNA damage**
- C9. M/ metaphase_checkpoint checks for **proper attachment** of all chromosomes to kinetochore **microtubules/ spindle fibres** OR proper alignment of all chromosomes at the metaphase plate/equator
- C10. When there is DNA damage, the p53 gene is activated to produce p53 proteins to upregulate the expression of some genes
- C11. Cell cycle is halted
- C12. so that there is enough time for cell to repair its damaged DNA
- C13. When DNA damage is beyond repair, apoptosis or cell death is initiated to stop the production of mutant daughter cells
- C14. Length of telomeres limit the number of cell divisions (Cells with telomeres at critical length do not go through any further cell divisions)
- C15. Presence/expression of telomerase in **some** cells (e.g. stem cells) to lengthen telomeres

Population level (max 4)

- P1. Genetic stability at population level means maintaining same frequency of alleles in the population;
- P2. Genetic stability occurs when there is **no change** in allele frequency, hence no major evolutionary forces such as natural selection, mutation, gene flow, genetic drift and sexual selection. **[mention at least 2]**

Events that contribute to genetic stability in a population (**P2-P8 - max 2**)

- P3. NO natural selection: individuals with favourable alleles will NOT passed them on to viable, fertile offspring hence allele frequency will NOT increase and contribute to genetic stability.
- P4. Constant unchanging environment leads to no change in selection pressure.
- P5. NO gene flow: there is NO transfer of alleles from one population to another through eg: migration
- P6. NO genetic drift: there is NO a change in allele frequency due to chance events on random individuals which tends to reduce genetic stability in populations.
- P7. Random mating: In sexually-reproducing populations, individuals DO NOT preferentially choose mates thus there is random fusion of gametes and genetic stability occurs. So random mating contributes to genetic stability, whereas non-random mating does not.
- P8. NO mutation, new alleles are NOT generated and genetic stability occurs.
- P9. Ref to binary fission that occurs in prokaryotic cells to produce daughter cells containing identical copy of the parental genome
- P10. Ref to absence of meiosis/crossing over in prokaryotes that ensures **no** new combinations of alleles/variation in genome
- P11. Ref to asexually reproducing populations/prokaryotes where there is no fusion of (genetically-different) gametes
- P12. Meiosis produces nuclei with half the number of chromosomes of parental nuclei (reductive division). This allows the diploid state to be restored upon fusion of gametes during fertilisation;

AVP

QWC: at least one correct marking pt about importance of genetic stability (**N**) + at least one correct mark point from each of the two categories (**M**olecular, **C**ellular, **P**opulation).

QUESTION 5

The cell is the basic unit of life, within which numerous biological processes that are essential to life occur.

(a) Discuss the importance of forming bonds that are weak and temporary during biological processes in the cell. [12]

1. *ref. to dynamic needs within cell/ such bonds only required transiently to allow transient process (or OWTTE)*
2. *ref. to named bond, e.g. hydrogen bond*
3. *ref. to such bonds being extensive in some cases, hence facilitating processes involving binding*

Weak and temporary bonds are involved in:

(enzymes)

4. weak bonds (or e.g. hydrogen bonds, ionic bonds) are formed between **substrate** and **active site** of enzyme
5. *ref. to **contact/ binding residues** holding substrates weakly in the correct orientation (or OWTTE) for catalysis*
6. such that **product** (when formed) can be readily released (or OWTTE)
7. *ref. to **competitive inhibitor** binding reversibly to **active site** OR (reversible) **non-competitive inhibitor** binding to **allosteric site***
8. *ref. to reversible binding not damaging enzyme structure (or OWTTE)*
9. *ref. to **end-product inhibition**, where end-product binds reversibly to allosteric site on earlier enzyme (or OWTTE) in **metabolic pathway***
10. *ref. to **regulation** is possible, since inhibitory effect is lifted when end-product concentration drops (or OWTTE)*

(max 4)

(DNA replication/protein synthesis)

11. **complementary base pairing**
12. during **DNA replication**, **hydrogen bonds** are formed between incoming **DNA nucleotides** and the **DNA template** strand
13. during **transcription**, **hydrogen bonds** are formed between incoming **RNA nucleotides** and the **DNA template** strand to **synthesise mRNA**
14. *ref. to weak hydrogen bonds being easily broken/ disrupted so that **DNA molecule**/ both **DNA strands** are easily separated to allow for DNA replication/ transcription*
15. during **translation**, **hydrogen bonds** are formed between **anti-codon** of **tRNA** and the (complementary) **codon** on **mRNA**
16. *ref. to complementary base pairing allowing for specific nucleotide/ amino acid sequences to be formed*
17. *ref. to genetic information is accurately expressed/ inherited*
18. *ref. to maintaining secondary/tertiary structure of proteins for function*

(max 5)

(cellular transport processes)

19. ligand binds weakly to **binding site** of **carrier protein**

- 20. carrier proteins are able to **change conformation** due to **weak bonds** (or e.g. hydrogen bonds, ionic bonds) **between R groups** that maintain **tertiary structure/ 3D conformation**
- 21. ligand binds weakly to **binding site** of receptor involved in **receptor-mediated endocytosis**
- 22. *ref. to* ligand being able to dissociate from carrier protein/ receptor after transport (*or OWTTE*)
- 23. **oxygen** (gas) binds weakly to **haem** group of **haemoglobin**
- 24. such that can be readily released where oxygen concentration is low
- 25. *ref. to* **subunits** of haemoglobin being held together by weak bonds (or e.g. hydrogen bonds, ionic bonds), hence allow for **co-operative binding** of oxygen

(max 4)

(other examples)

- 26. e.g. **ligand** binds reversibly to **receptor** on cell surface during **cell signaling**
- 27. *ref to* weak bonds between ligand and receptor to allow for **ligand dissociation** for signal termination/after cellular response has been carried out
- 28. e.g. weak bonds are formed between **homologous chromosomes** during **synapsis**/ to form **bivalents**
- 29. e.g. weak bonds involved in **cell-cell recognition/ adhesion**
- 30. e.g. weak bonds are formed during **antibody-antigen interaction**

31. AVP

QWC: importance of (some) named weak and temporary bonds communicated clearly and unambiguously to include **at least three** different examples (roles) in cellular processes (*not structure*)

(b) Cells are able to divide and give rise to new cells.

Cyclins are an important family of proteins that control the rate of cell cycle progression of a dividing cell. They are so named because, in general, the concentrations of the various cyclins change in a cyclical manner during the cell cycle. The changing levels of active cyclins thus ensure that there is sufficient time for various checks to be performed, to ensure that key cell cycle events have occurred normally.

Outline the checks made at cell cycle checkpoints, and explain how a dividing cell is able to **quickly** regulate the levels of cyclins which in turn control these checkpoints. [13]

Part I: outline the checks made at cell cycle checkpoints

1. The three checkpoints are the **G1**, **G2** and **M/ metaphase** checkpoints
2. at the G1 checkpoint, **sufficient nutrients/ growth factors** are checked for
3. and whether there is **adequate cell size/ duplication of all organelles/ no DNA damage** before the cell proceeds to the S phase
4. at the G2 checkpoint, **DNA replication** is assessed for **completion**
5. and that there is **no DNA damage/ cell size is adequate** (award once only)
6. at the M/ metaphase checkpoint, **proper attachment of all chromosomes** to kinetochore **microtubules/ spindle fibres** is checked for
7. and all chromosomes are properly **aligned** at the **metaphase plate/ equator**, before the dividing cell is allowed to enter anaphase
8. *ref. to cell cycle checkpoints allowing cell cycle to proceed only when internal and external conditions are favourable (or OWTTE)*
9. *ref. to providing time for errors/ malfunctioning cellular machinery to be repaired/ prevent errors in cell division that may result if the cell cycle progressed prematurely to the next stage (or OWTTE)*

[max 5]

*Part II: explain how a dividing cell is able to **quickly** regulate the levels of cyclins*

(From contextual hints of “concentrations of the various cyclins change in a cyclical manner during the cell cycle” and “quickly regulate”: Expression of the various cyclins is controlled at the transcriptional, post-transcriptional, translational and post-translational levels.)

(A: transcriptional level control)

10. cyclin concentrations can be increased/ decreased through increasing/ decreasing the **rate of gene transcription**
11. **specific transcription factors** bind to **control elements**
12. **activator** protein binds to **enhancer** sequence on DNA
13. example of how bound activators function, e.g. **binding with GTFs/ mediator proteins** to **promote assembly** of the **transcription initiation complex**, recruiting **histone remodelling complexes**
14. **repressor** protein binds to **silencer** sequence on DNA
15. example of how bound repressors function, e.g. **competing with activator** for **overlapping binding sites**, **binding** to (activation domain of) **activator** hence preventing its function, **binding** to **GTF** hence **preventing assembly of TIC**, recruiting **histone deacetylases/ histone methyltransferases**

16. *ref. to* **controlling/ stabilising/ destabilising** the **rate of formation of transcription initiation complex** at the **promoter/ assembly** of **RNA polymerase** and **general transcription factors** at the **promoter** to form transcription initiation complex

[max 4]

(B: *post-transcriptional level control*)

17. addition of **5' cap** and **3' poly(A) tail** to **mRNA** occurs
 18. which **prevents** the mRNA from being (immediately) **degraded** by nucleases in the cytoplasm
 19. **regulates nuclear export** of mRNA
 20. promotes translation/ facilitates binding of translation initiation factors/ facilitates ribosome attachment

[max 3]

(C: *translational level control*)

21. the **longer** the **poly(A) tail**, the **longer** the **half-life** of mRNA / longer the time taken for the poly(A) tail to be **shortened to critical length** resulting in complete breakdown of mRNA
 22. *ref. to* **more stable mRNA** leading to **increased rate of translation** to produce more cyclins
 23. cyclin concentrations can be increased/ decreased through increasing/ decreasing **rate of translation**
 24. **translational repressor proteins** bind to mRNA
 25. **preventing** the **binding** of small and large **ribosomal subunits** to **mRNA** at its 5' UTR
 26. thus **preventing** the **formation of translation initiation complex** (*award only once*)
 27. **activation/ inactivation** of **translation initiation factors** by dephosphorylation phosphorylation
 28. **promotes/ prevents** the **formation of translation initiation complex** (*award only once*)

[max 4]

(D: *post-translational level control*)

29. cyclin proteins can be **activated** by **chemical modification/ phosphorylation**
 30. concentrations of active cyclins can be decreased through **protein degradation**
 31. unwanted cyclins are **tagged with ubiquitin/ undergo ubiquitination**
 32. and are subsequently hydrolysed/ degraded within **proteasomes**

[max 3]

33. AVP

34. QWC: good spread of knowledge communicated without ambiguity to include:

- from Part I – at least 1 marking point
- from Part II – at least 1 marking point each from any 2 sections (A, B, C or D)